CiPoNT Patch

(Copper ions,PLGA,NSAIDs,TXA)

Domain - Human Wellbeing and Wellness

Aim: To provide the complete solution against animal bites and scratches and the related infections.

59,000 people die due to scratches and bites every year.

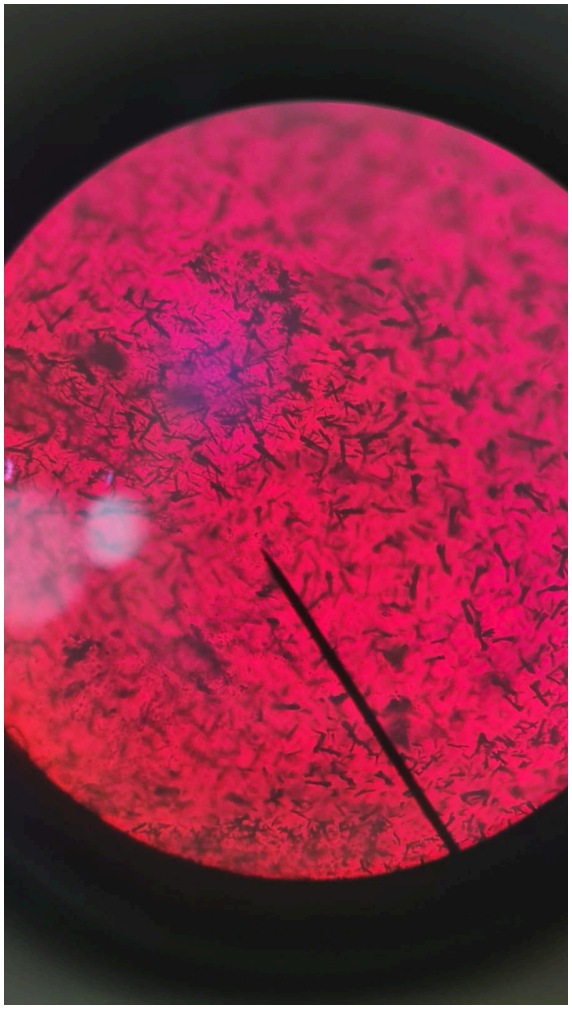
Such an injury poses three problems,
Pain - Bleeding - Infection

Pain and Bleeding resolved by NSAIDs and TXA
Infection controlled by Copper ions (vector PLGA through microneedle patch)

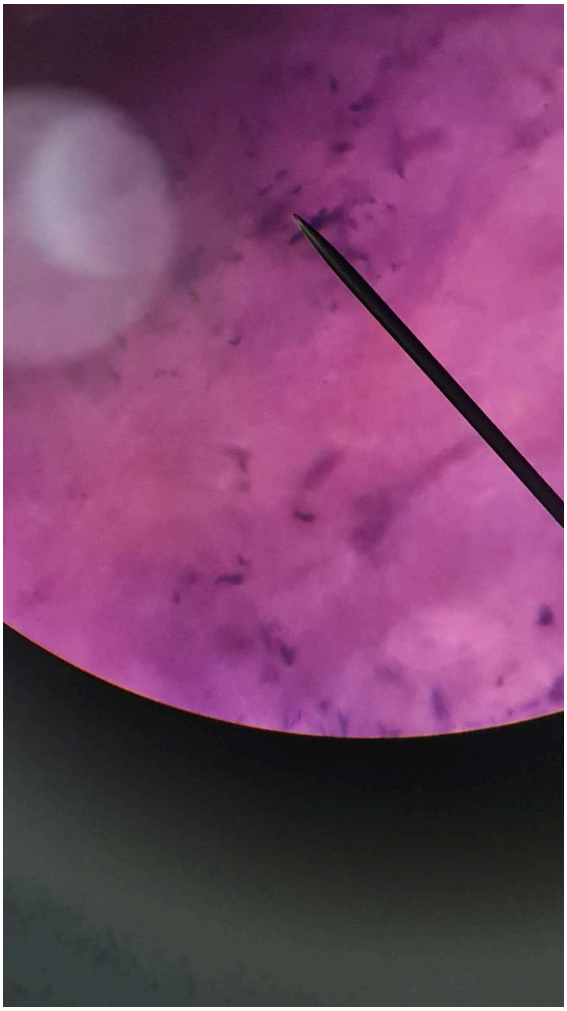
Microneedle Patch - Made of hydrogel, hard on dry surface allowing for penetration and after penetration takes up skin interstitial fluid to expand resulting in largening of holes to deliver PLGA (having copper ions) to render virus ineffective
PLGA[1] - Carriers for controlled CuNP[2] release
Copper ions - Create oxidative stress, which can disassemble bacterial or viral membranes
NSAIDs[3] - Common non-steroidal drugs administered to reduce pain and inflammation
TXA[4] - Tranexamic Acid to promote blood clotting preventing blood loss
Microneedle Patch and Drug reservoir are separate with pellets in (40-20-10-5) kg doses to be used as required

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[1] Poly(lactide-co-glycolide)
[2] Copper Nanoparticles
[3] Non-steroidal Anti Inflammatory Drugs
[4] Tranexamic Acid



WITHOUT COPPER IONS



WITH COPPER IONS

Microneedle Patch separate CiNaNT Drugs separate
Pellets of different size - (40,20,10,5) kg per dose

40 - 2.8 l blood
20 - 1.4 l blood
10 - 0.7 l blood
5 - 0.35 l blood

Copper nanoparticle
Concentration : 5mg/liter of blood
Weight :
Cost: 0.0025$ per
Proof of Concept - Done (Bacillus bacteria from Cat)
How does it work: ROS Reactions, membrane stress, breakdown of inner mechanism

Tranexamic Acid
Weight:
Cost:

NSAIDS
Weight :
Cost:

poly(lactide-co-glycolide) (PLGA).

Drug Carriers

Polymer Structure

Surface binding rebinding release mechanism

Ph dependent release

Carries CuNP through surface mechanisms

Weight :

Cost:

1. Gender:
    - Adult males: ~5.5 liters
    - Adult females: ~4.5 liters
2. Body weight:
    - ~7-8% of total body weight in adults
    - ~8-9% in children
    - ~9-10% in infants
3. Pregnancy: Pregnant women have 50-60% more blood volume compared to non-pregnant women.
4. Age:
    - Infants: ~75-85 ml per kg of body weight
    - Children: ~70-75 ml per kg of body weight
    - Adults: ~65-70 ml per kg of body weight

67*40 = 2.6*5 = 13 mg ————>40 kg

Pellet composition

Body weight (Kg)	Blood volume (L)	Dose of CuNp (mg)	Dose of TXA (mg)	Dose of Ipubrufen (mg)	PLGA molecule (Carriers) (mg)
40	2.8	14	600	300	87.5
20	1.4	7	300	150	43.75
10	0.7	3.5	150	75	21.88
5	0.35	1.75	75	37.5	10.94

8mg Cu in 50 mg PLGA